

EXP:28

Find the boundary of the image using Convolution kernel for the given image.

AIM:- The boundary of the image using Convolution kernel for the given image.

PROGRAM:-

```
import cv2

import numpy as np

# --- Step 1: Load the image ---
img = cv2.imread(r'C:\Users\pavan\OneDrive\OpenCV\x.jpg')
if img is None:
    print("Error: Image not found or path is incorrect")
    exit()

# --- Step 2: Convert to grayscale ---
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

# --- Step 3: Define convolution kernels ---
# Prewitt operator kernels
kernel_x = np.array([[ -1, 0, 1],
                     [ -1, 0, 1],
                     [ -1, 0, 1]])

kernel_y = np.array([[ -1, -1, -1],
                     [ 0, 0, 0],
                     [ 1, 1, 1]])

# --- Step 4: Apply convolution ---
edge_x = cv2.filter2D(gray, -1, kernel_x)
```

```
edge_y = cv2.filter2D(gray, -1, kernel_y)
```

```
# --- Step 5: Combine results ---
```

```
edges = cv2.addWeighted(edge_x, 0.5, edge_y, 0.5, 0)
```

```
# --- Step 6: Show and save results ---
```

```
cv2.imshow("Original Image", img)
```

```
cv2.imshow("Boundary Detected Image", edges)
```

```
cv2.imwrite("Boundary_Image.jpg", edges)
```

```
cv2.waitKey(0)
```

cv2.destroyAllWindows()

INPUT:



OUTPUT:



RESULT:- Thus the program is executed successfully.